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Diseñe la armadura de la viga 25/50 mostrada en la figura .Considere hormigón G25, acero A630-420H y un recubrimiento de 5 cm. Datos: $P_L := 9.3\ \text{tonnef}$ $q_L := 800\ \frac{\text{kgf}}{\text{m}}$

Nota : No olvide considerar el peso propio de la viga

Viga simplemente apoyada con carga distribuida y una carga puntual al centro de la luz

Datos

$b := 25\ \text{cm}$ $h := 50\ \text{cm}$ $L := 6\ \text{m}$ $\gamma_c := 2500\ \frac{\text{kgf}}{\text{m}^3}$

$r := 5\ \text{cm}$ $d := h - r = 45\ \text{cm}$

$f_c := 250\ \frac{\text{kgf}}{\text{cm}^2}$ $f_y := 4200\ \frac{\text{kgf}}{\text{cm}^2}$ $\beta := 0.85$ $E_s := 2100000\ \frac{\text{kgf}}{\text{cm}^2}$

Solución

$q_{pp} := \gamma_c \cdot b \cdot h = 0.3125\ \frac{\text{tonnef}}{\text{m}}$

$q_D := q_{pp}$

$M_{u1} := \frac{1}{8} \cdot 1.4 \cdot q_D \cdot L^2 = 1.9688\ \text{tonnef m}$

$M_{u2} := \frac{1}{8} \cdot 1.2 \cdot q_D \cdot L^2 + \frac{1}{8} \cdot 1.6 \cdot q_L \cdot L^2 + 1.6 \cdot P_L \cdot \frac{L}{4} = 29.7675\ \text{tonnef m}$

$M_u := \max\left(\begin{bmatrix} M_{u1} \\ M_{u2} \end{bmatrix}\right) = 29.7675\ \text{m tonnef}$

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Condición de diseño

$\phi M_n := M_u \qquad \phi := 0.9$

$\phi M_n := \phi \cdot \rho \cdot b \cdot d^2 \cdot f_y \cdot \left(1 - 0.59 \cdot \rho \cdot \frac{f_y}{f_c}\right)$

$$\rho := \text{polyroots}\left(\begin{bmatrix} -M_u \\ \phi \cdot b \cdot d^2 \cdot f_y \\ (-\phi) \cdot b \cdot d^2 \cdot f_y \cdot 0.59 \cdot \frac{f_y}{f_c} \end{bmatrix}\right) = \begin{bmatrix} 0.0192 \\ 0.0817 \end{bmatrix}$$

$\rho := 0.192$

$\rho_{max} := 0.32 \cdot \beta \cdot \frac{f_c}{f_y} = 0.0162 \qquad \rho_{max} < \rho$ no cumple la norma

$A_{smax} := \rho_{max} \cdot b \cdot d = 18.2143\ \text{cm}^2$

$M_{n1} := \rho_{max} \cdot b \cdot d^2 \cdot f_y \cdot \left(1 - 0.59 \cdot \rho_{max} \cdot \frac{f_y}{f_c}\right) = 28.9005\ \text{m tonnef}$

$\phi\left(M_{n1} + M_{n2}\right) := M_u$

$M_{n2} := \frac{M_u}{\phi} - M_{n1} = 4.1745\ \text{m tonnef}$

Suponer que A's fluye

$M_{n2_} := A'_s \cdot f_y \cdot (d - d')$

$d' := r = 5\ \text{cm}$

$A'_s := \frac{M_{n2}}{f_y \cdot (d - d')} = 2.4848\ \text{cm}^2$ 4 barras de 10 mm

Equilibrio de fuerzas

$0.85 \cdot f_c \cdot a \cdot b + A'_s \cdot f_y := A_s \cdot f_y$

$a := \frac{A_{smax} \cdot f_y}{0.85 \cdot f_c \cdot b} = 14.4\ \text{cm} \qquad c := \frac{a}{\beta} = 16.9412\ \text{cm}$

$\varepsilon_c := 0.003$

$\varepsilon'_s := \frac{\varepsilon_c}{c} \cdot (c - d') = 0.0021 \qquad \varepsilon'_s > \varepsilon_y$ Fila de aceros en compresión fluye ==>
Supuesto se cumple

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$A_s := A_{smax} + A'_s = 20.6991\ cm^2$

Armadura inferior

Seis barras de diametro 18 ($A_s := 22.81\ cm^2$)

$n_{filas} := 2$ $n_b := 6$ $\phi_b := 22\ mm$ $\phi_s := 10\ mm$ $r_{est} := 3\ cm$ (al borde libre lateral del estribo)

$$s := \frac{b - 2\cdot r_{est} - \frac{n_b}{n_{filas}}\cdot\phi_b - 2\cdot\phi_s}{\frac{n_b}{n_{filas}} - 1} = 5.2\ cm \qquad s_{min} := \max\left(\begin{bmatrix} 25\ mm \\ \phi_b \end{bmatrix}\right) = 2.5\ cm$$

$s > s_{max}$ ok

$d_{eq} := d - \frac{\phi_b}{2} - \frac{25}{2}\ mm = 42.65\ cm$

$A'_s := A_s - A_{smax} = 4.5957\ cm^2$

Armadura superior

Cuatro barras de diametro 12 ($A'_s := 4.52\ cm^2$)

$n_{filas} := 1$ $n_b := 4$ $\phi_b := 12\ mm$ $\phi_s := 10\ mm$ $r_{est} := 3\ cm$ (al borde libre del estribo)

$$s := \frac{b - 2\cdot r_{est} - \frac{n_b}{n_{filas}}\cdot\phi_b - 2\cdot\phi_s}{\frac{n_b}{n_{filas}} - 1} = 4.0667\ cm \qquad s_{min} := \max\left(\begin{bmatrix} 25\ mm \\ \phi_b \end{bmatrix}\right) = 2.5\ cm$$

$s > s_{min}$ ok

$$M_n := \rho_{max}\cdot b\cdot d_{eq}^{\ 2}\cdot f_y\cdot\left(1 - 0.59\cdot\rho_{max}\cdot\frac{f_y}{f_c}\right) + A'_s\cdot f_y\cdot\left(d_{eq} - d'\right) = 33.1083\ tonnef\ m$$

$\phi\cdot M_n = 29.7974\ tonnef\ m$

$FU := \frac{M_u}{\phi\cdot M_n} = 0.999$ ok

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